

Springer Book Information Form

(1) Title of the book

Field	Answer
Title	From Molecules to Quantum Circuits
Subtitle	A Computational Guide to Fermion-to-Qubit Encodings

(2) Author's / editor's affiliation

Field	Author 1 — Affiliation 1
Name	John S
Surname	Azariah
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Field	Author 1 — Affiliation 2
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Add additional authors in a second table if needed.

(3) Author's / editor's profile (~300 words)

John Azariah is a software engineer, language designer, and quantum computing researcher with over 30 years of experience in professional software development at Microsoft. His career spans foundational contributions to Microsoft applications like Office, Microsoft tools and languages like J#, and Microsoft Azure Services like Azure Batch, Azure Kubernetes Service (AKS), and Azure Storage. He served as compiler lead and language designer behind Microsoft Q#, Microsoft's quantum programming language, before releasing Azure Quantum as well.

John holds a BA in Computer Science and Chemistry from Dartmouth College and is currently a PhD candidate in Quantum Computing at the University of Technology Sydney under the supervision of Dr Chris Ferrie at the Centre for Quantum Software and Information.

He is the creator of the FockMap library, an open-source F# framework for symbolic Fock-space operator algebra that serves as the computational backbone of this book. FockMap provides a

composable, type-safe pipeline for constructing qubit Hamiltonians, performing qubit tapering, and compiling Trotter circuits — with zero runtime dependencies on any quantum computing platform.

John’s unique combination of academic grounding in Chemistry and Computer Science; deep industry software engineering expertise; and active academic research in quantum computing positions him to bridge the gap between theoretical quantum chemistry and practical circuit compilation. He maintains a technical blog at <https://johnazariah.github.io> covering topics in quantum computing, functional programming, and software engineering. His pedagogical approach of bridging the gap between relatable problems and discovering underlying structure reflects a commitment to getting to the bottom of things so his readers can understand the *why* while applying it to the *what*.

(4) About the book (~300 words + chapter-wise description)

This book is a self-contained, 23-chapter computational tutorial covering the complete pipeline from molecular electronic structure to quantum circuit compilation for quantum simulation. It addresses a critical gap in the literature: no existing resource walks a student through every coefficient, every sign, and every intermediate Pauli string while providing executable code that reproduces each result.

Starting from one-body and two-body integrals of the hydrogen molecule (H₂) in the STO-3G basis, we construct the qubit Hamiltonian explicitly under six fermion-to-qubit encodings (Jordan–Wigner, Bravyi–Kitaev, Parity, balanced binary tree, balanced ternary tree, and a Vlasov complete-ternary-tree encoding), verify spectral equivalence across encodings, reduce qubit count via diagonal and Clifford Z₂ symmetry tapering, decompose the tapered Hamiltonian into Trotter circuits with explicit CNOT gate counts, and export the result to OpenQASM 3.0 and Q#. Two running examples — H₂ (4 qubits) and H₂O (12–14 qubits) — are developed from molecular geometry to quantum circuit, including computing the H₂ dissociation curve and the H–O–H bond angle from first principles.

Every formula has a corresponding executable computation in the companion FockMap library, an open-source F# framework for symbolic Fock-space operator algebra. The book includes 10 interactive laboratory sessions.

Chapter-wise description:

Stage	Chapters	Description
The Molecule	Ch 1–3	Electronic structure, mathematical notation, spin-orbital formalism
The Machine	Ch 4	Qubits, gates, and circuits
Encoding	Ch 5–9	Visual introduction to encodings, Hamiltonian construction, six encoding schemes, Vlasov tree, spectral verification
Tapering	Ch 10–13	Motivation, diagonal Z ₂ symmetries, Clifford rotation tapering, benchmarks
Circuits	Ch 14–17	Time evolution, Trotter decomposition, CNOT staircase compilation, gate cost analysis

Stage	Chapters	Description
The Pipeline	Ch 18–21	End-to-end pipeline, H ₂ O bond angle, VQE/QPE algorithms, OpenQASM 3.0 / Q# export
Horizons	Ch 22–23	Scaling analysis, future directions
Appendices	A–B	API cookbook, mathematical background

(5) Unique selling points (USPs)

- **Complete pipeline in one place:** The only resource that covers the entire path from molecular integrals to compiled quantum circuits, with no steps treated as a black box.
- **Six encodings side by side:** Comprehensive comparison of Jordan–Wigner, Bravyi–Kitaev, Parity, binary tree, ternary tree, and Vlasov encodings — most resources cover only one or two.
- **Every intermediate result shown:** Every coefficient, sign, and Pauli string is derived explicitly, enabling readers to verify each step.
- **Executable companion code:** All formulas are backed by runnable F# scripts using the open-source FockMap library — readers can reproduce every result.
- **Two realistic molecules:** H₂ and H₂O are developed end-to-end, including computing physically meaningful results (dissociation curve, bond angle).
- **Platform-agnostic:** Unlike tutorials tied to Qiskit or Cirq, this book exports to both OpenQASM 3.0 and Q#, focusing on the mathematics rather than a specific SDK.
- **Hands-on lab sessions:** 10 interactive F# lab scripts guide the reader through building encodings from scratch.
- **Bridges three communities:** Accessible to quantum computing researchers, computational chemists, and software engineers alike, with prerequisites clearly stated and all necessary background developed in-text.

(6) Type of work

Monograph

(7) Originality of the work

Option	Applicable?
New	Yes
Previously published with Springer	No
Previously published with other publisher	No
Previously published electronically	No
Translated	No
Revised	No

Note: The content is original and has not been peer-reviewed or published by any academic publisher.

(8) Main discipline(s) / sub-discipline(s) / MSC codes

#	Discipline
1	Quantum Computing
2	Quantum Chemistry / Quantum Simulation
3	Computational Physics
4	Mathematical Physics
5	Computer Science — Formal Methods
6	Physical Chemistry — Electronic Structure
7	Applied Mathematics
8	Scientific Computing

MSC 2010 codes:

Code	Description
81P68	Quantum computation
81V55	Molecular physics
81-01	Instructional exposition (quantum theory)
81Q05	Closed and approximate solutions to the Schrödinger equation
15A66	Clifford algebras, spinors
68Q12	Quantum algorithms and complexity
92E10	Molecular structure

(9) Courses (for textbooks)

#	Course	Typical Institution
1	Quantum Computing	UTS, MIT, ETH Zürich, Caltech
2	Quantum Chemistry Simulation	University of Toronto, TU Delft
3	Quantum Information Science	Stanford, University of Waterloo
4	Computational Quantum Physics	University of Oxford, MPQ Munich
5	Introduction to Quantum Algorithms	University of Maryland, UCL
6	Quantum Software Engineering	University of Technology Sydney
7	Advanced Topics in Quantum Computing	Harvard, University of Chicago

(10) Content level

Level	Applicable?
Lower undergraduate	
Upper undergraduate	Secondary
Graduate	Primary
Professional	Secondary
Practitioner	
Research	Secondary
Popular	
General	

(11) Estimated number of

Metric	Count
Chapters	23 + 2 appendices (25 total)
Manuscript pages	~166
Tables	
Illustrations / Images	27 (25 diagrams + 2 data plots)

(12) Keywords (minimum 6)

1. fermion-to-qubit encoding
2. quantum chemistry simulation
3. Jordan–Wigner transform
4. Bravyi–Kitaev transform
5. qubit tapering
6. Trotterization
7. quantum circuit compilation
8. molecular Hamiltonian
9. Vlasov encoding
10. quantum computing tutorial

(13) Author’s / editor’s earlier books

Title	Publisher	ISBN	Year

(14) Competition (Springer / non-Springer)

Book	Publisher	Year	Why readers would choose <i>our</i> book
Nielsen & Chuang, <i>Quantum Compu- tation and Quantum Informa- tion</i>	Cambridge University Press	2010	Covers QC foundations but not the chemistry-to-circuit pipeline; no executable code
Szabo & Ostlund, <i>Modern Quantum Chem- istry</i>	Dover	1996	Develops electronic structure theory but stops before qubit encoding
McArdle et al., <i>Quantum Compu- tational Chem- istry</i> (review)	Rev. Mod. Phys.	2020	Algorithmic overview without step-by-step derivations or runnable code
Helgaker, Jør- gensen & Olsen, <i>Molecu- lar Electronic- Structure Theory</i>	Wiley	2000	Comprehensive reference but no quantum computing content
Hidary, <i>Quantum Comput- ing: An Applied Approach</i>	Springer	2021	Broad QC textbook; encodings covered only briefly

Book	Publisher	Year	Why readers would choose <i>our</i> book
Schuld & Petruccione, <i>Supervised Learning with Quantum Computers</i>	Springer	2018	ML-focused; does not cover quantum chemistry simulation pipeline

Key differentiator: No competing title covers the *complete* pipeline from molecular integrals through six encodings, tapering, Trotterization, and circuit export — with every intermediate result shown and executable.

(15) Manuscript submission date

A complete first draft of the manuscript is available (166 pages, ~47,000 words, 23 chapters + 2 appendices). The author is considering adding material on qubitization — covering block encoding and the linear combination of unitaries (LCU) framework — to both the companion FockMap library and the book. Reformatting for Springer style and any requested revisions would constitute the remaining work.

Field	Answer
Estimated delivery	

(16) Experts in the topic

#	Name	Affiliation	Email
1	Dr Guang Hao Low	Google Quantum AI	Confirmation requested
2	Dr Nathan Wiebe	Pacific Northwest National Laboratory (PNNL) / University of Toronto	Confirmation requested
3	Dr Stephen Jordan	Google Quantum AI	Confirmation requested